TABLEAU ASSIGNMENT

Parameters, Filters, and Parameter Data Types

1) What is a Parameter in Tableau and how does it differ from a filter?

A **Parameter** in Tableau is a user-defined value that can be changed to control calculations, reference lines, or other parts of a visualization. A parameter can hold a value such as a number, date, text, or Boolean value. It is often used with a calculated field to let users interact with a dashboard.

A **Filter** is used to include or exclude specific data from a view, such as showing only the East region or sales from a particular year. A filter directly restricts the data displayed, while a parameter is an independent input value and does not change the view by itself. To make a parameter affect the data, it usually needs to be used in a calculation, filter condition, or another supported feature.

Example: A Region filter can show only the East region. A parameter can let a user enter a target sales value, which a calculated field then uses to label sales as above or below target.

2) List three use cases where Parameters are preferred over filters.

1. Dynamic measure selection: Let users switch the chart between Sales, Profit, and Quantity using one control and a calculated field.

2. What-if analysis or target setting: Allow users to change a target value, discount rate, or forecast assumption and see how the results change.

3. Top N analysis: Let users choose how many top products or customers to display, such as the Top 5, Top 10, or Top 20, using a parameter in a calculation or filter logic.

3) What are the four data types supported in Tableau parameters?

The four commonly supported parameter data types are:

Data Type	Description	Example
Integer	Whole numbers without a decimal part.	10
Float	Numbers that can include decimal places.	12.5
String	Text values.	"East"
Date	Calendar date values.	2026-10-29

Note: Tableau's available parameter data types can vary by version and context. In current Tableau versions, parameter data types commonly include Integer, Float, Boolean, String, Date, and Date & Time. If your course asks for four, it may be referring to a simplified or older list; check the exact list taught in class.

Conclusion

Parameters provide flexible user-controlled inputs for interactive analysis, while filters directly restrict which data appears in a view. Parameters are especially useful for dynamic measure selection, what-if analysis, and Top N analysis.